Regularized Discrete Optimal Transport

S. Ferradans, N. Papadakis, G. Peyré and J.F. Aujol, 2014

Statistical Optimal Transport : Article Review

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April 22nd 2026

Contents

Introduction

We present here a short review of the article "Regularized Discrete Optimal Transport" [1] by S. Ferradans, N. Papadakis, G. Peyré and J.F. Aujol, published in *SIAM Journal in Imaging Sciences* in 2014. The authors introduce a relaxed and regularized formulation of the optimal transport (OT) problem and different resolution algorithms, as well as applications in color transfer between pictures in high resolution.

OT methods have been used to solve image processing tasks since the late 1990s, with first applications in computer vision. Rubner et al. [2] introduced in 1998 the use of Earth Mover's Distance (EMD) to compare visual signatures (ie. "histograms" of colors, similar to distributions) between pictures, analogous to the Wasserstein W_1 metric in the case of unequal masses between 2 signatures. In the following years, Haker et al. [3] adopted continuous formulations of OT (now using the Wasserstein W_2 metric) for image registration in medical imaging and morphing. Color transfer tasks were first studied by Pitié et al. [4] reducing the 3D projection problem onto the RGB space to a 1D OT problem, laying the groundwork for the sliced Wasserstein distance. In 2013, Marco Cuturi's introduction of Sinkhorn distances [5] allowed crucial developments of the OT algorithms, enhancing their speed and making the problem differentiable, a significant advance for image processing and for the entire field.

The fundamental concepts of relaxation and regularization of the Kantorovich OT problem discussed by the authors were already broadly studied prior to this publication. Firstly, relaxing the mass conservation constraint in transport amounts to loosening the one-to-one mapping constraint in graph matching, a topic adressed by Zaslavskiy, Bach and Vert [6], in order to avoid the NP-hard quadratic assignment problem. Secondly, such a relaxation is a necessary condition for solving certain regularized problems as shown by Louet and Santambrogio [7] for Sobolev regularizations. Finally, defining the regularization term itself isn't trivial and requires results from manifold learning, following in the footsteps of Elmoataz et al. [8] studying nonlocal discrete regularization.

The rest of this review is structured as follows: in Section 1, we present the discrete formulation of optimal transport and its relaxed extension; in Section 2, we introduce the discrete regularized optimal transport framework together with the associated optimization schemes; in Section 3, we describe applications to color processing tasks such as color transfer and color normalization; finally, in Section 4, we discuss the limitations of the approach and review subsequent developments that addressed these issues.

1 Problem Presentation: From Discrete Monge to Relaxed Optimal Transport

We consider two point clouds $X = \{x_i\}_{i=1}^N$ and $Y = \{y_j\}_{j=1}^N$ in $\mathbb{R}^d$. We define the associated empirical measures with uniform weights:

$$\mu_X = \frac{1}{N}\sum_{i=1}^N \delta_{x_i}, \quad \mu_Y = \frac{1}{N}\sum_{j=1}^N \delta_{y_j}. \tag{1}$$

The Monge problem consists in finding a transport map T pushing μ_X onto μ_Y verifying the equation $T_\# \mu_X = \mu_Y$ while minimizing a given cost function. In the discrete setting with uniform weights, this reduces to finding a permutation $\sigma \in \Sigma_N$:

$$\min_{\sigma\in\Sigma_N} \frac{1}{N}\sum_{i=1}^N C_{i,\sigma(i)}. \tag{2}$$

Equivalently, this can be written as matrix minimization problem on the set of permutation matrices $\mathcal{P}$:

$$\min_{\Sigma\in\mathcal{P}} \langle C_{X,Y}, \Sigma\rangle \tag{3}$$

$$\mathcal{P} = \left\{\Sigma \in \{0,1\}^{N\times N} \mid \Sigma\mathbf{1} = \mathbf{1},\ \Sigma^T\mathbf{1} = \mathbf{1}\right\}. \tag{4}$$

The Kantorovich relaxation replaces the discrete constraint $\Sigma_{i,j} \in \{0,1\}$ with a convex one replacing $\mathcal{P}$ by the set of bistochastic matrices. By Birkhoff's theorem described by Marshall et al. [9], the solution remains a permutation matrix, which explains the rigidity of the model.

$$\mathcal{S}_1 = \left\{\Sigma \in [0,1]^{N\times N} \mid \Sigma\mathbf{1} = \mathbf{1},\ \Sigma^T\mathbf{1} = \mathbf{1}\right\}. \tag{5}$$

To overcome this rigidity, a second relaxation is introduced by loosening the marginal constraints. We define the set $\mathcal{S}_\kappa$:

$$\mathcal{S}_\kappa = \left\{\Sigma \in [0,1]^{N\times N} \,\middle|\, k_X\mathbf{1} \le \Sigma\mathbf{1} \le K_X\mathbf{1},\ k_Y\mathbf{1} \le \Sigma^T\mathbf{1} \le K_Y\mathbf{1},\ \mathbf{1}^T\Sigma\mathbf{1} = M\right\}. \tag{6}$$

We can note that this new relaxation weakens the marginal constraints, allowing more flexible transport patterns. A source point x_i can distribute its mass across multiple targets y_j, while a target point y_j can receive mass from several sources. As a result, the transport plan Σ is no longer restricted to a one-to-one correspondence and instead exhibits a more diffuse behavior.
To ensure $\mathcal{S}_\kappa$ is non-empty, M must satisfy:

$$\frac{M}{N} \in [\min(k_X, k_Y), \max(K_X, K_Y)]. \tag{7}$$

A relaxed transport map is defined[1] from Σ as a weighted barycentric projection of Y :

$$T(X) = (\mathrm{diag}(\Sigma \mathbf{1}))^{-1} \Sigma Y. \tag{8}$$

The author states that if (k_X, K_X, k_Y, K_Y, M) are integers, then there exists a solution $\tilde{\Sigma}$ of the relaxed problem such that $\tilde{\Sigma}_{i,j} \in \{0, 1\}$. [Appendix 4, Fig 1]

2 Discrete Regularized Optimal Transport

To incorporate geometric structure into optimal transport, we define a weighted graph on each point cloud. For a dataset X, we consider $G_X = (X, E_X, W_X)$. X are the vertices, E_X is the set of edges constructed using a k-nearest neighbors rule, and W_X denotes the associated edge weights. An edge (i, j) is included if x_j belongs to the k nearest neighbors of x_i, and weights are typically defined as $w_{i,j} = \|x_i - x_j\|^{-1}$. On this graph, we define a discrete gradient operator acting on vector-valued functions V as:

$$(G_X V)_{i,j} = w_{i,j}(V_i - V_j). \tag{9}$$

G_X helps to measure variations of a function with respect to the underlying geometry of the data. Based on this construction, we introduce a (p, q)-norm graph regularization:

$$J_{p,q}(G_X V) = \sum_{(i,j)\in E_X} \|w_{i,j}(V_i - V_j)\|_q^p. \tag{10}$$

The operator J penalizes irregular variations of V along the graph and enforces consistency of the transport with the geometry of the data manifold.

We now embed this structure into a regularized optimal transport framework between two point clouds X and Y. Let $\Sigma \in \mathcal{S}_\kappa$ denote a transport plan. The associated objective function reads:

$$f(\Sigma) = \langle C_{X,Y}, \Sigma \rangle + \frac{\lambda_X}{2} \|\Gamma_{X,Y}(\Sigma)\|^2 + \frac{\lambda_Y}{2} \|\Gamma_{Y,X}(\Sigma)\|^2, \tag{11}$$

The graph-structured operators are defined by:

$$\Gamma_{X,Y}(\Sigma) = G_X \Delta_{X,Y}(\Sigma), \quad \Gamma_{Y,X}(\Sigma) = G_Y \Delta_{Y,X}(\Sigma). \tag{12}$$

The optimization problem being convex and constrained over the transport polytope $\mathcal{S}_\kappa$ makes it suitable for projection-free methods. We therefore rely on a Frank-Wolfe scheme, where each iteration consists of solving a linearized subproblem followed by a convex combination update[2]:

$$\tilde{\Sigma}^{(\ell+1)} = \arg\min_{\tilde{\Sigma}\in\mathcal{S}_\kappa} \langle \nabla f(\Sigma^{(\ell)}), \tilde{\Sigma} \rangle, \tag{13}$$

$$\Sigma^{(\ell+1)} = \Sigma^{(\ell)} + \tau_\ell (\tilde{\Sigma}^{(\ell+1)} - \Sigma^{(\ell)}), \tag{14}$$

[1]The condition $k_X > 0$ ensures the map is well-defined.

[2]$\tau_\ell \in [0, 1]$ is chosen via line search.

The gradient of the objective combines the transport cost and the graph-based regularization terms. Its explicit expression, together with the definition of the associated adjoint operators, is provided in Appendix 4. Each Frank-Wolfe iteration thus reduces to solving a linear minimization problem over $\mathcal{S}_\kappa$, which can be efficiently handled using standard linear programming techniques.

When $(p, q) = (1, 1)$, the regularization becomes non-smooth and corresponds to an anisotropic total variation penalty. Introducing auxiliary variables U_X, U_Y, the problem can be reformulated as a linear program. Replacing the non-differentiable regularization by linear constraints, enable the use of standard linear programming solvers.

$$\min_{\Sigma, U_X, U_Y} \langle C_{X,Y}, \Sigma \rangle + \lambda_X \langle U_X, \mathbf{1} \rangle + \lambda_Y \langle U_Y, \mathbf{1} \rangle, \tag{15}$$

subject to:

$$\begin{cases} -U_X \leq G_X(\Sigma Y - \operatorname{diag}(\Sigma \mathbf{1}) X) \leq U_X, \\ -U_Y \leq G_Y(\Sigma^T X - \operatorname{diag}(\Sigma^T \mathbf{1}) Y) \leq U_Y, \\ \Sigma \in \mathcal{S}_\kappa. \end{cases} \tag{16}$$

The role of λ_X and λ_Y is to control the trade-off between fidelity to the transport cost and geometric regularity. Increasing these parameters enforces consistency of the transport with the graph structures of X and Y. In particular, the regularization promotes alignment of geometric structures rather than pointwise correspondences, leading to transport plans that respect the clustering structure of the data [Fig 2].

3 Applications to Color Processing

The proposed framework can be applied to color processing [Fig 3] tasks such as color transfer and color normalization. Given an input image X^0 and a target distribution, the goal is to modify the colors of X^0 so that they match a desired palette. Relaxing the mass conservation constraint allows better matching of dominant color modes, while regularization reduces noise amplification and visual artifacts.

To make the method tractable for high-resolution images, the approach relies on a three-step pipeline. First, the color distributions are down-sampled by clustering the images into a smaller set of representative points using K-means. Then, a regularized transport map T is computed between these reduced point clouds by solving the relaxed optimal transport problem. Finally, the map is extended to the full-resolution image using a nearest-neighbor interpolation.The up-sampling step restores fine-scale details and improves visual quality.

$$T^0(x) = T(X_{i(x)}) + x - X_{i(x)}, \quad \text{where } i(x) = \arg\min_{1 \leq i \leq N} \|x - X_i\|. \tag{17}$$

In the case of multiple input images, this framework naturally extends using regularized OT barycenters 4. Instead of choosing a single reference image, a common color palette is computed as the barycenter of all input distributions. The barycenter minimizes a weighted sum of regularized transport distances and can be efficiently approximated using a block-coordinate descent scheme. Each image is then transported toward this shared palette using the same pipeline as above leading to consistent color normalization across the dataset, while avoiding the bias induced by selecting a single reference image.

4 Limits and further developments

The relaxed and regularized formulation of discrete OT by Ferradans et al. had significant impact in color transfer applications : regularization allowed for strong noise reduction during transfer, while the problem relaxation helped solve the issue of unequal masses between color signatures. In this framework they also managed to compute a barycenter of input distributions and discussed the applications to color normalization. As the authors mentioned, there are some limitations to their approach and further works expanded on these issues, as the field of OT strongly developed in the following years.

First, the Sobolev or TV regularization on a K-nearest neighbors graph requires solving linear or quadratic programming problems with heavy computations. To ease the weight of computations, the authors significantly reduce the image resolution of their inputs (with $N = 400$ clusters usually chosen as a good resolution-computational cost compromise) and retrieve the final million-pixel output through heuristic interpolations, resulting in quantization errors. To avoid solving this complex 3D problem, Bonneel et al. [10] reduce it to multiple 1D projections, simplifying the OT problem with the sliced Wasserstein distance and avoiding K-means subsampling of the input image.

Moreover, although introducing the "box" constraints on the permutation matrix space allows for a relaxed solution between unequal masses, the k_X and K_X constants are difficult to calibrate and interpret theoretically. Chizat et al. [11] introduced the more robust formulation of unbalanced OT, with flexible penalties based on statistical divergences (like Kullback-Leibler) to allow for mass creation or destruction. This type of approach has fully replaced box constraints in relaxed OT frameworks. Another innovation on the spatial regularization was introduced by Solomon et al. [12]: to avoid the costly K-nn graph construction, authors showed that a Sinkhorn regularization was equivalent to applying gaussian filters or convolutional Wasserstein distances, expanding on Marco Cuturi's results to increase resolution speed.

Finally, using a purely discrete OT plan has become obsolete with recent development in deep learning. Input convex neural networks were used to directly approximate the continuous Monge transport map rather than a discrete approximation [13], expanding on Brenier's theorem to model continuous spatial OT and color transfer.

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Appendix

Binary Optimal Solutions via Total Unimodularity

Proposition 1. *Let* $(k_X, K_X, k_Y, K_Y, M) \in (\mathbb{N}^*)^5$. *Then the relaxed problem6 admits at least one optimal solution* $\tilde{\Sigma}$ *such that*

$$\tilde{\Sigma} \in \{0, 1\}^{N \times N}.$$

Proof. Let $\mathcal{S}_\kappa$ denote the feasible set of the relaxed problem. By construction, it can be written as

$$\mathcal{S}_\kappa = \left\{ \Sigma \in \mathbb{R}^{N \times N} \;\middle|\; \mathcal{A}(\Sigma) \leq b_\kappa \right\},$$

where $\mathcal{A}$ is the linear map encoding the row and column marginal constraints together with the global mass constraint

$$\mathcal{A}(\Sigma) = \left(-\Sigma,\ \Sigma\mathbf{1},\ -\Sigma\mathbf{1},\ \Sigma^*\mathbf{1},\ -\Sigma^*\mathbf{1},\ \mathbf{1}^\top\Sigma\mathbf{1},\ -\mathbf{1}^\top\Sigma\mathbf{1}\right).$$

, and where

$$b_\kappa = (0_N,\ K_X\mathbf{1},\ -k_X\mathbf{1},\ K_Y\mathbf{1},\ -k_Y\mathbf{1},\ M,\ -M).$$

The constraint matrix associated with $\mathcal{A}$ is totally unimodular. Since b_κ has integer components, every vertex of the polytope $\mathcal{S}_\kappa$ is integral. Moreover, because the relaxed problem includes the bounds $0 \leq \Sigma_{ij} \leq 1$, every integral feasible point of $\mathcal{S}_\kappa$ must belong to $\{0, 1\}^{N \times N}$.

Finally, the objective function of 6 is linear, so an optimal solution can be chosen at an extreme point of the feasible polytope. Therefore, there exists an optimal solution $\tilde{\Sigma}$ that is binary. □

Detailed Expressions for the Optimization Scheme

This appendix provides the explicit expressions used in the Frank-Wolfe optimization scheme described in the main text.

The objective function combines the transport cost with graph-based regularization terms. Its gradient with respect to Σ is given by:

$$\nabla f(\Sigma) = C_{X,Y} + \lambda_X \Delta^*_{X,Y}\big(G^*_X \Gamma_{X,Y}(\Sigma)\big) + \lambda_Y \Delta^*_{Y,X}\big(G^*_Y \Gamma_{Y,X}(\Sigma)\big). \tag{18}$$

The adjoint operator $\Delta^*_{X,Y}$ is explicitly defined by:

$$\Delta^*_{X,Y}(U) = \mathrm{diag}(UX^T)\mathbf{1}^T - UY^T, \tag{19}$$

and $\Delta^*_{Y,X}$ is defined analogously.

These expressions are obtained by direct differentiation of the graph-regularized transport functional and allow efficient computation of first-order information.

Given the search direction

$$E^{(\ell)} = \tilde{\Sigma}^{(\ell+1)} - \Sigma^{(\ell)}, \tag{20}$$

the step size minimizing the quadratic surrogate along this direction admits a closed-form expression:

$$\tau_\ell = \frac{-\langle E^{(\ell)}, C_{X,Y}\rangle - \lambda_X \langle \Gamma_{X,Y}(E^{(\ell)}), \Gamma_{X,Y}(\Sigma^{(\ell)})\rangle - \lambda_Y \langle \Gamma_{Y,X}(E^{(\ell)}), \Gamma_{Y,X}(\Sigma^{(\ell)})\rangle}{\lambda_X \|\Gamma_{X,Y}(E^{(\ell)})\|^2 + \lambda_Y \|\Gamma_{Y,X}(E^{(\ell)})\|^2}. \tag{21}$$

Representation of Relaxed and Regularizes Transport

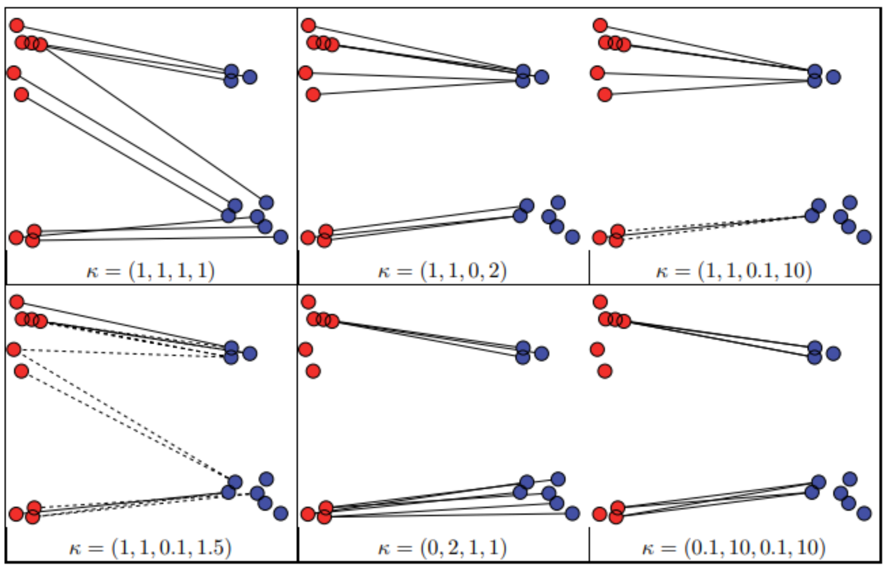

Figure 1: Relaxed optimal transport between X (red points) and Y (blue points) for different values of κ. The case $\kappa = (1, 1, 1, 1)$ corresponds to classical optimal transport. The transport plan $\Sigma_{i,j}$ defining correspondences between X_i and Y_j is visualized by line segments, drawn as dashed when $\Sigma_{i,j} \in (0.1, 1)$ and solid when $\Sigma_{i,j} = 1$.

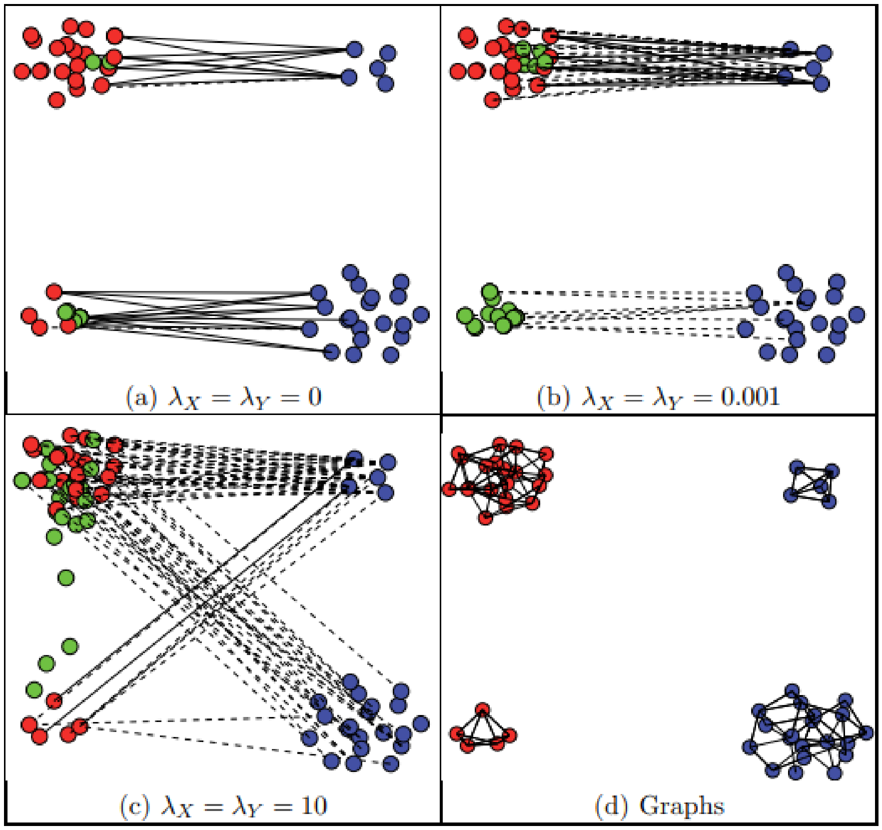

Figure 2: Given two point sets X (blue) and Y (red), we represent the transported points $Z = \mathrm{diag}(\Sigma\mathbf{1})^{-1}\Sigma Y$ (green), together with the transport mappings $\Sigma_{i,j}$ between X_i and Y_j. These correspondences are illustrated by line segments, drawn as dashed when $\Sigma_{i,j} \in (0.1, 1)$ and solid when $\Sigma_{i,j} = 1$. The results are obtained using the relaxed and regularized optimal transport formulation with parameters $\kappa = (0.1, 8, 0.1, 8)$.

Regularized Optimal Transport Barycenters

For advanced imaging applications like texture mixing or color normalization, it is highly useful to compute a distribution that acts as the "average" (barycenter) of a set of input distributions. To simplify the optimization process, an asymmetric version of the regularized OT energy is utilized. By maintaining one data set X as the reference and applying regularization only with respect to its own graph (setting $\lambda_Y = 0$), the distance simplifies to:

$$D(\mu_X, \mu_Y) = \min_{\Sigma \in \mathcal{D}_\kappa} \langle C_{X,Y}, \Sigma \rangle + \lambda J_{p,q}(G_X \Delta_{X,Y}(\Sigma)). \tag{22}$$

with

$$\mathcal{D}_\kappa = \left\{ \Sigma \in [0,1]^{N \times N} \;\middle|\; \Sigma^T \mathbf{1} \leq k\mathbf{1},\ \Sigma\mathbf{1} = 1 \right\}. \tag{23}$$

Given a set of input clouds $(X^{[r]})_{r \in \mathcal{R}}$ and their respective weights $(\rho_r)_{r \in \mathcal{R}}$, the barycenter cloud X is defined as a local minimizer of the weighted sum of these distances:

$$\min_{X \in \mathbb{R}^{N \times d}} E_\rho(X) = \sum_{r \in \mathcal{R}} \rho_r D(\mu_{X^{[r]}}, \mu_X). \tag{24}$$

While the energy is not jointly convex, it is separately convex with respect to the barycenter X and the individual transport plans $(\Sigma^{[r]})_{r \in \mathcal{R}}$. Therefore, the solution can be approximated using a block-coordinate descent scheme, which alternates between two steps. First, we update $\Sigma^{[r]}$. Fixing the barycenter X, the algorithm performs $|\mathcal{R}|$ independent regularized OT optimizations in parallel to update the transport matrices. Then we update X. Fixing the transport plans, the algorithm updates the positions of the barycenter points. For Sobolev regularization ($p = q = 2$), this reduces to an unconstrained quadratic problem solved efficiently via a symmetric linear system (e.g., using a conjugate gradient algorithm). For TV regularization ($p = q = 1$), the non-smooth problem can be solved using a first-order primal-dual splitting scheme.

Color Transfer Representation

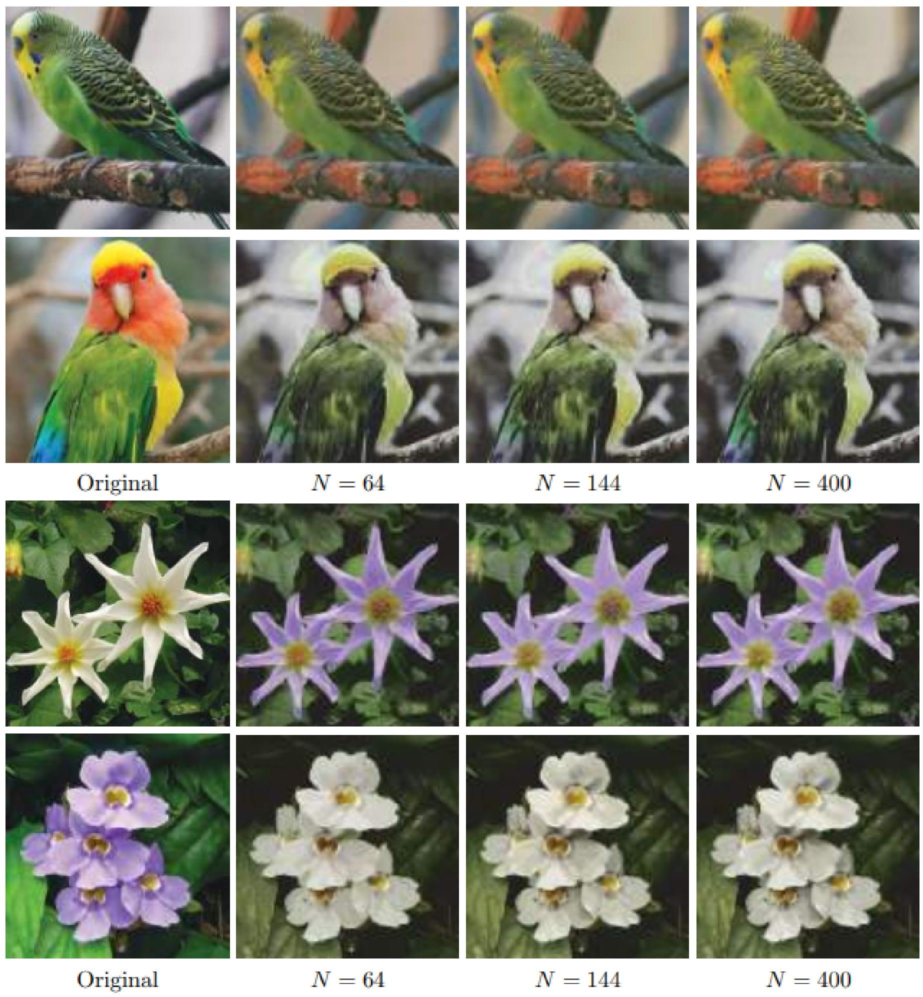

Figure 3: Results obtained by computing relaxed and regularized optimal transport between the original images for varying values of N. The number of points is fixed to $n = 4$, with $M = N$ in all experiments. For the first and second rows, the parameters are set to $\kappa = (0.1, 1, 0.1, 1)$ and $\lambda_X = \lambda_Y = 10^{-3}$. For the third and fourth rows, we use $\kappa = (0.1, 1.1, 0.1, 1.1)$ and $\lambda_X = \lambda_Y = 9 \cdot 10^{-4}$. Each column corresponds to a separate experiment with a different value of N.